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IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR

AIM:

To design and configure multiple subnets using Cisco Packet Tracer, assign static IP addresses to hosts in each subnet, connect the subnets through a router, and simulate packet transmission to study communication between different subnets.

ALGORITHM:

1. Open Cisco Packet Tracer and create a new network topology.
2. Select the required PCs, switches, and a router from the device library.
3. Divide the network into multiple subnets according to the number of hosts required.
4. Assign suitable IP addresses and subnet masks to each subnet.
5. Connect the PCs to their respective switches using Copper Straight-Through cables.
6. Connect the switches to the router using Ethernet cables.
7. Configure the router interfaces with suitable IP addresses for each subnet.
8. Enable the router interfaces and configure the default gateway on each PC.
9. Use the ping command to test communication between PCs in the same subnet.
10. Test communication between PCs belonging to different subnets.
11. Observe packet transmission using Simulation mode in Cisco Packet Tracer.
12. Disconnect or disable the router interfaces and repeat the ping test to observe the change in communication.

PROGRAMS / CONFIGURATION:

No separate program is required. The network is designed and configured using Cisco Packet Tracer.

IP ADDRESS CONFIGURATION:

Subnet 1:

PC0 IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Router Interface G0/0: 192.168.1.1

Subnet 2:

PC1 IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Router Interface G0/1: 192.168.2.1

ROUTER CONFIGURATION:

```
enable
 configure terminal
 interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit
 interface gigabitEthernet 0/1
 ip address 192.168.2.1 255.255.255.0
 no shutdown
 exit
 end
```

Note: Interface names may vary depending on the router model selected in Packet Tracer.

COMMAND:

ping 192.168.2.2

OUTPUT:

Case 1: When the subnets are connected through a router.

The ping command succeeds, and packets are transmitted between the PCs in different subnets when the IP addresses, default gateways, and router interfaces are configured correctly.

Case 2: When the router is disconnected or its required interfaces are disabled.

The ping command between different subnets fails because there is no functioning route between the networks.

RESULT:

Thus, subnetting was implemented and simulated successfully using Cisco Packet Tracer. Static IP addresses and subnet masks were assigned, and communication between different subnets was tested through a router. The experiment demonstrated the importance of routers in enabling communication between separate IP networks.